

The Coupled Gated Tensor-Hidden Semi-Markov Model (CGT-HSMM)

A Unified Bayesian Predictor for Latent Regime Dynamics under Noisy
State-Dependent Multi-Source Annotations

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Abstract

This paper introduces the Coupled Gated Tensor-Hidden Semi-Markov Model (CGT-HSMM), a unified Bayesian framework that integrates multi-source annotator calibration with temporal sequence estimation. Traditional crowdsourcing models (e.g., Dawid-Skene, HMM-Crowd) assume either independent observations or static annotator accuracy. In contrast, the CGT-HSMM models the latent states as a discrete-time (monthly) Hidden Semi-Markov process while conditioning the annotator emission distributions on the active latent state using a state-dependent 3-way tensor CP decomposition. We prove oracle-regime emission identifiability under Kruskal rank constraints, delineate precisely which components of full-model identifiability are established in the literature (the geometric-dwell special case) and which remain conjectural for general dwell laws, and derive a score-autocovariance effective-sample-size correction for the marginal-score (Godambe) estimator, used as a conservative operational gate. Furthermore, we audit the framework adversarially, presenting critical challenges and defenses, and we state acknowledged limitations explicitly.

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1 Introduction

Modern decision-support systems in public policy, epidemiology, and geopolitical risk tracking rely on noisy, heterogeneous data channels. To aggregate these channels, statisticians employ latent class models (like the Dawid-Skene model) to calibrate source reliability. Separately, to capture the temporal progression of regimes, researchers employ sequence models (such as Hidden Markov Models).

However, in real-world scenarios, these two processes are coupled:

1. **State-Dependent Reliability:** A source’s reliability (sensitivity and false-positive rate) is rarely static; it shifts depending on the underlying latent state. For instance, media feeds, institutional report streams, scraper pipelines, or separately modeled annotation channels often experience elevated false-positive rates during volatile policy regimes. Coders reading the same document are annotation channels, not independent acquisition sources; they enter the likelihood through a separate Dawid-Skene coder-reliability layer.
2. **Uncertainty Propagation:** Decoupled pipelines estimate source parameters first and plug them into sequence estimators as constants. This underestimates the posterior variance, leading to overconfident forecasts.

To address these limitations, we present the **Coupled Gated Tensor-Hidden Semi-Markov Model (CGT-HSMM)**. This model integrates state-dependent source calibration and sequence dynamics into a single, unified Bayesian likelihood. Additionally, it implements a stack of fail-closed mathematical gates that abstain from forecasting when parameters are unidentifiable, uninformative, or prior-dominated.

2 Mathematical Formulation of the CGT-HSMM

Let $Z_t \in \{1, \dots, L\}$ be a discrete-time (monthly) Hidden Semi-Markov process representing the latent regime at time $t \in \{1, \dots, T\}$. Let T_n denote the index of the n -th regime transition. The sequence of states $S_n = Z_{T_n}$ forms the embedded Markov chain. The model is discrete-time throughout; “continuous-time HSMM” language used in earlier drafts is withdrawn as it is inconsistent with the shifted-Poisson dwell distributions and the monthly observation grid.

2.1 Latent Semi-Markov Dynamics

The latent transition behavior is governed by two components:

1. **Embedded Transition Matrix:** Let $\Pi \in \mathbb{R}^{L \times L}$ be the transition probability matrix of the embedded chain, with zero diagonal:

$$\pi_{ii} = 0 \quad \forall i \in \{1, \dots, L\}$$

enforcing that transitions must represent a change of state.

2. **Dwell-Time Distribution:** Let the dwell time D_r in state r follow a discrete probability mass function $f_r(d; \theta_r)$ with finite mean $\mu_r = E[D_r] < \infty$.

2.2 State-Dependent Tensor Emissions

At each discrete time step $t \in \{1, \dots, T\}$, we observe a binary vector of annotations $\mathbf{Y}_t = (Y_{1,t}, \dots, Y_{S,t})^T \in \{0, 1\}^S$ from S conditionally independent sources.

Unlike standard models where annotator accuracy is constant, the emission distribution is conditioned on the active latent state $Z_t = r$. For each state r , the emission is modeled as a latent mixture representing the sub-regimes of the policy stratum:

$$P(\mathbf{Y}_t \mid Z_t = r) = \sum_{a \in \{0,1\}} \pi_a(r) \prod_{s=1}^S (M_{r,a}^{(s)})^{Y_{s,t}} (1 - M_{r,a}^{(s)})^{1-Y_{s,t}} \quad (1)$$

where:

- $\pi_1(r) = P(A_t = 1 \mid Z_t = r)$ is the sub-regime prevalence prior in state r , and $\pi_0(r) = 1 - \pi_1(r)$.
- $M_{r,a}^{(s)}$ represents the emission parameters of independent acquisition channel s conditioned on sub-regime a and latent state r :

$$M_{r,1}^{(s)} = P(Y_{s,t} = 1 \mid A_t = 1, Z_t = r) \quad (\text{Sensitivity under state } r)$$

$$M_{r,0}^{(s)} = P(Y_{s,t} = 1 \mid A_t = 0, Z_t = r) \quad (\text{False Positive Rate under state } r)$$

When the channels are coders reading the *same* item, they do not satisfy the conditional-independence assumption (A2) and must instead be modeled in a separate Dawid–Skene coder-reliability layer; they do not count toward the $S \geq 3$ Kruskal source condition.

Equation (1) represents a state-dependent Canonical Polyadic (CP) tensor decomposition of the joint observation space for each latent state r .

2.3 Joint Posterior Density

Let $\Theta = \{\Pi, \{\boldsymbol{\theta}_r\}_{r=1}^L, \{\pi_a(r), \mathbf{M}_r\}_{r=1}^L\}$ be the set of all model parameters. The joint posterior density of the parameters and the latent trajectory $Z_{1:T}$ is:

$$P(\Theta, Z_{1:T} \mid \mathbf{Y}_{1:T}) \propto P(\Theta) \left(P(Z_{1:T} \mid \Pi, \boldsymbol{\theta}) \prod_{t=1}^T P(\mathbf{Y}_t \mid Z_t, \mathbf{M}, \boldsymbol{\pi}) \right) \quad (2)$$

where $P(\Theta)$ represents the prior distribution of the parameters.

3 Parameter Identifiability: Proven and Conjectured Components

Because the regime sequence $Z_{1:T}$ and the sub-regime states $A_{1:T}$ are unobserved, the parameters Θ must be recovered from the joint law of the observations $P(\mathbf{Y}_{1:T})$. An earlier draft of this section claimed full identifiability of Θ via a three-stage argument whose first stage conditioned on the latent state $Z_t = r$ as if it were observable; since the per-state tensor is not a functional of the observed law, that argument was circular, and it is withdrawn (see the closing remark). This section states precisely what is proven and what is conjectured.

3.1 Proposition 1: Oracle-Regime Emission Identifiability (proven)

Theorem 1 (Oracle-regime emission identifiability). *Fix a regime r and suppose the regime path $Z_{1:T}$ were observed. Conditional on the path, the observations $\{\mathbf{Y}_t : Z_t = r\}$ are i.i.d.*

draws from the 2-class product-Bernoulli latent class model with prevalence $\pi_1(r)$ and factor matrices

$$M_r^{(s)} = \begin{pmatrix} 1 - M_{r,0}^{(s)} & 1 - M_{r,1}^{(s)} \\ M_{r,0}^{(s)} & M_{r,1}^{(s)} \end{pmatrix}, \quad \det M_r^{(s)} = M_{r,1}^{(s)} - M_{r,0}^{(s)},$$

and the visit set $\{t : Z_t = r\}$ grows linearly in T by positive recurrence, supplying the required replication. If $S \geq 3$, $M_{r,1}^{(s)} \neq M_{r,0}^{(s)}$ for all s , and $\pi_1(r) \in (0, 1)$, then the Kruskal condition $\sum_s k(M_r^{(s)}) = 6 \geq 2R + 2$ holds and $\{\pi_a(r), \mathbf{M}_r\}$ is identifiable up to the sub-regime label swap — exactly as in Theorem A' of the companion document, whose proof (including positivity of the weights, scaling resolution by column-stochasticity, and the covariance-based exclusion of degenerate alternatives) transfers verbatim.

3.2 Proposition 2: Dynamics Identifiability Given the Regime Path (proven, elementary)

Theorem 2 (Oracle-path dynamics identifiability). *If $Z_{1:T}$ were observed, the embedded transition matrix Π and the dwell laws f_r are identified in the limit $T \rightarrow \infty$ by the empirical transition frequencies of the embedded chain and the empirical dwell histograms, by the strong law of large numbers under irreducibility and positive recurrence.*

3.3 Full Identifiability from the Observed Law: Status

The substantive question is whether Θ is identified from $P(\mathbf{Y}_{1:T})$ *without* the oracle path. We record the status honestly.

1. **Geometric dwell (HMM special case): established in the literature.** If all f_r are geometric, the model is a stationary finite HMM with state-conditional emission laws $Q_r = P(\mathbf{Y}_t \mid Z_t = r)$ on $\{0, 1\}^S$. By the HMM identifiability theorem of Allman–Matias–Rhodes (2009, Thm. 6), the HMM parameters are generically identifiable from the joint law of finitely many consecutive observations provided Π is full rank and the emission laws $\{Q_r\}_{r=1}^L$ are linearly independent as vectors in $\mathbb{R}^{2^S}$. The within-state mixture parameters $\{\pi_a(r), \mathbf{M}_r\}$ are then recovered from each identified Q_r by Proposition 1's Kruskal argument.
2. **General dwell: conjectured, with a named obstruction.** An HSMM with dwell support bounded by $d_{\max}$ embeds as an HMM on the augmented state (Z_t, a_t) with at most $L \cdot d_{\max}$ states. The standard spectral conditions *fail as stated* for this embedding: emission laws are shared across ages of the same regime, so the augmented emission family $\{Q_{(r,a)}\}$ contains repeated members and is never linearly independent. Identification must instead exploit longer observation windows that separate regimes through their dwell signatures. We *conjecture* that Θ is identifiable under (i) full-rank Π , (ii) linearly independent regime emission laws $\{Q_r\}$, and (iii) dwell laws with distinct identifiable probability-generating functions and aperiodic lattice support (distinct finite means are a convenient sufficient diagnostic for distinctness but not a complete condition; two dwell laws can agree in mean yet be separated by higher moments, and conversely distinct means do not guarantee linear independence of the regime emission laws); we do not prove it. The conjecture is probed operationally by simulation-based calibration, which would expose non-identifiability as multimodal or non-converging posteriors across simulated worlds.
3. **Label anchoring (downstream of identifiability).** Whatever identification holds is up to a permutation of regime labels. The production implementation anchors on the ordered first-tactic activity vector (the `ordered` declaration in `m01_forward.stan`); distinct dwell means provide an alternative anchor. Anchoring resolves the label permutation; it cannot create identifiability that is otherwise absent.

Remark 1 (Withdrawn claims from the earlier draft). *Three steps of the earlier “proof” are withdrawn: (i) Stage 1 applied Kruskal’s theorem to the per-state tensor $(T_r)_{ijk} = P(\mathbf{Y}_t = \cdot \mid Z_t = r)$, which is not observable — the observed law is the mixture over r , and disaggregating it is the very thing to be proven; (ii) Stage 2 cited “the path uniqueness of semi-Markov processes (Allman et al., 2009)” — no such result appears in that reference; (iii) Stage 3 estimated Π and f_r from latent transition times that are not data, which presupposes Stage 2. Propositions 1–2 retain the salvageable content of Stages 1 and 3 under an explicit oracle hypothesis; the unproven gap is isolated in the conjecture above.*

4 Effective Sample Size under Dependent Observations

We analyze how the information about the sub-regime prevalence $\pi_1(r)$ scales when observations are not independent but are coupled to the latent transition structure. The object controlled below is the asymptotic variance of the *marginal-score* (composite-likelihood) estimator. $I_F^{(\text{marg})}$ is the Godambe information of the marginal-score estimator and is used as a conservative operational diagnostic; we do not claim it universally bounds the full-likelihood Fisher information of the HSMM (that claim would require additional regularity and nuisance-parameter conditions not verified here). The resulting gates are conservative in their operational design stance.

4.1 Theorem 2: Effective Sample Size under Temporal Dependency

Theorem 3 (Dependent-Sample Information Scaling). *Let $Y_{1:N}$ be a stationary binary observation sequence from the channel model with informativeness $I_r = |\rho_r - \delta_r|$ and marginal success probability p_r , driven by a stationary latent activity process A_t with summable autocovariances and satisfying the positivity condition $\gamma_0 + 2 \sum_{h \geq 1} \gamma_h > 0$ (required for the asymptotic variance to be finite). Let s_t be the per-observation marginal score for $\pi_1(r)$, with autocovariances $\gamma_h = \text{Cov}(s_0, s_h)$. Then the estimator solving the marginal score equation $\sum_t s_t = 0$ has asymptotic variance $(N_{\text{eff}} I_r^2 / (p_r(1 - p_r)))^{-1}$, where*

$$N_{\text{eff}} = N \frac{\gamma_0}{\gamma_0 + 2 \sum_{h \geq 1} \gamma_h}. \quad (3)$$

If, in addition, A_t is a binary first-order Markov chain with non-unit eigenvalue $\lambda \in [0, 1)$, then $\text{Corr}(A_0, A_h) = \lambda^h$ exactly and

$$N_{\text{eff}} = \frac{N}{1 + 2c \frac{\lambda}{1 - \lambda}}, \quad c = \frac{I_r^2 \pi_1(r)(1 - \pi_1(r))}{p_r(1 - p_r)} \in (0, 1], \quad (4)$$

and the textbook expression $N(1 - \lambda)/(1 + \lambda)$ (the specialization $c = 1$) is a conservative lower bound on N_{eff} . For the CGT-HSMM itself the activity process is not first-order Markov, and (4) is a labeled approximation only; see the remark following the proof.

4.2 Step-by-Step Proof

Proof. Step 1 (sandwich variance). The marginal score is $s_t = I_r(Y_t - p_r)/(p_r(1 - p_r))$, a stationary mean-zero sequence with $\gamma_0 = I_r^2/(p_r(1 - p_r))$. For the M-estimator solving $\sum_{t=1}^N s_t(\hat{\pi}) = 0$, standard sandwich asymptotics give $\sqrt{N}(\hat{\pi} - \pi) \Rightarrow \mathcal{N}(0, V/H^2)$ with $H = \gamma_0$ (the expected negative score derivative, which here equals the marginal information) and long-run score variance

$$V = \lim_{N \rightarrow \infty} \frac{1}{N} \text{Var} \left(\sum_{t=1}^N s_t \right) = \gamma_0 + 2 \sum_{h \geq 1} \gamma_h,$$

where the limit uses the Cesàro convergence $\lim_N [\gamma_0 + 2 \sum_{h=1}^{N-1} (1 - h/N) \gamma_h] = \gamma_0 + 2 \sum_{h \geq 1} \gamma_h$ under summability. The asymptotic variance is therefore $V/(NH^2) = (N_{\text{eff}} \gamma_0)^{-1}$ with N_{eff} as in (3).

Step 2 (score autocovariance reduces to activity autocovariance). Since $P(Y_t = 1 | A_t) = \delta_r + I_r A_t$ and the channel noise is independent across t given the activity path, for $h \geq 1$ we have $\text{Cov}(Y_0, Y_h) = I_r^2 \text{Cov}(A_0, A_h)$, hence

$$\frac{\gamma_h}{\gamma_0} = c \text{Corr}(A_0, A_h), \quad c = \frac{I_r^2 \pi_1(r)(1 - \pi_1(r))}{p_r(1 - p_r)},$$

and $c \leq 1$ by the law of total variance, $p_r(1 - p_r) = E[\text{Var}(Y | A)] + I_r^2 \pi_1(r)(1 - \pi_1(r))$, with equality only for a noiseless channel.

Step 3 (Markov corollary). If A_t is a two-state Markov chain, its autocorrelation is exactly λ^h , where λ is the single non-unit eigenvalue of its transition matrix (this exactness is special to two states; for $L > 2$ the autocorrelation of an indicator is a mixture of eigenvalue powers). Substituting $\gamma_h = \gamma_0 c \lambda^h$ into (3) and summing the geometric series yields (4); setting $c = 1$ shrinks N_{eff} further, so $N(1 - \lambda)/(1 + \lambda) \leq N_{\text{eff}}$. $\square$

Remark 2 (Why the AR(1) form fails for the CGT-HSMM, and in which direction). *In the CGT-HSMM the activity is driven by the semi-Markov regime layer: the discrete-time regime process is a renewal-type process, not a Markov chain, so “the subdominant eigenvalue of the transition matrix of the latent states” is undefined, and $\text{Corr}(A_0, A_h)$ is a renewal functional of the dwell-time laws. Two failure modes of any single- λ fit follow. (i) For overdispersed dwell distributions, the autocorrelation decays sub-geometrically over the relevant lags; a geometric fit understates $\sum_h \gamma_h$ and overstates N_{eff} — anti-conservative, the dangerous direction. (ii) For strongly concentrated (quasi-periodic) dwell, the autocorrelation oscillates with negative lobes; $\sum_h \gamma_h$ shrinks or turns negative and the AR(1) form is over-conservative. Therefore (3) must be evaluated with $\sum_h \text{Corr}(A_0, A_h)$ computed from the fitted dwell and transition posteriors, and (4) may be reported only as a labeled approximation. The Godambe information of the marginal-score estimator is used as a conservative operational diagnostic. We do not claim it equals or universally bounds the full HSMM likelihood information. Gating on (3) is conservative by design stance, not by a general Fisher-information ordering theorem.*

5 Adversarial Mathematical Audit: Novelty & Critiques

To assess whether the CGT-HSMM represents a genuine mathematical contribution, we audit the framework against potential criticisms.

5.1 Is the Model Genuinely Novel?

- **Yes, in its coupling of states and errors:** Existing crowdsourcing models (e.g., Dawid-Skene, HMM-Crowd) treat annotator expertise as a static attribute of the coder. The CGT-HSMM models annotator sensitivity and false-positive rates as **state-dependent variables** ($\rho_s(Z_t), \delta_s(Z_t)$) that shift dynamically based on the latent discrete-time regime.
- **Contribution: the integration.** The contribution is the integration of state-dependent source calibration, semi-Markov latent regimes, and pre-registered abstention gates in one Bayesian architecture. Related ideas exist separately in latent-class modeling, HSMMs, selective prediction, decision-theoretic abstention, and conformal risk control; the claimed novelty is the specific coupling and gate stack, not the invention of each component. A systematic literature review of close prior art (abstaining classifiers, safe screening,

Bayesian decision rejection, robust multiple testing) is required before asserting novelty in any submission.

5.2 Theoretical Critique 1: The Curse of State-Dependent Parameter Inflation

Critique 1. *By conditioning the source parameters on the latent states, the number of parameters increases from $2S$ (in standard Dawid-Skene) to $2S \times L$ (in CGT-HSMM). In finite samples, this parameter inflation will lead to overfitting, high posterior variance, and a failure of the MCMC sampler to converge.*

Defense 1. *To mitigate parameter inflation, we introduce a **hierarchical shrinkage prior** on the source parameters. We model the state-dependent parameters as deviations from a global source profile:*

$$M_{r,a}^{(s)} \sim \text{Beta}(\alpha_{s,a} \cdot \kappa, \beta_{s,a} \cdot \kappa)$$

where the prior mean $\alpha_{s,a}/(\alpha_{s,a} + \beta_{s,a})$ encodes the global (state-free) profile of source s for sub-regime a , and κ is a concentration parameter setting the strength of shrinkage toward that profile.

5.3 Theoretical Critique 2: The Alignment and Label-Switching Problem

Critique 2. *Because MCMC samplers are invariant to permutations of the latent state labels, the model will suffer from label-switching during sampling. If the labels switch, the state-dependent source parameters $(\rho_s(r), \delta_s(r))$ will mix across chains, rendering the posterior draws uninterpretable and invalidating the gates.*

Defense 2. *To resolve the label-switching problem, we implement two constraints:*

1. **Ordering Constraint:** *We enforce an ordering constraint on the expected dwell times:*

$$\mu_1 < \mu_2 < \dots < \mu_L$$

*Since the states are ordered by their expected dwell, the labels are anchored, preventing permutation during sampling. (The production implementation anchors instead on the ordered first-tactic activity vector — the **ordered** declaration in `m01_forward.stan` — which serves the same purpose. Whichever anchor is chosen must be pre-registered, and any ordering constraint mildly distorts the posterior when the true ordered quantities are nearly tied.)*

2. **Identifiability Relabeling:** *During the post-processing of MCMC chains, we apply a relabeling algorithm (e.g., the Stephens algorithm) that minimizes the Kullback-Leibler divergence between the state-conditional emission distributions across draws, ensuring consistent parameter alignment.*

6 Conclusion

The CGT-HSMM couples state-dependent tensor emissions with semi-Markov dynamics, addressing the variance underestimation of decoupled calibrate-then-forecast pipelines. The adversarial audit yields a mixed but honest verdict: oracle-regime emission identifiability and the dependent-sample information correction are proven; full-model identifiability is established only in the geometric-dwell special case and is otherwise an explicitly stated conjecture probed by simulation-based calibration; parameter inflation and label switching are mitigated — not

eliminated — by hierarchical shrinkage and pre-registered anchoring. The framework’s scientific defensibility rests on this explicit separation of proven, conjectured, and mitigated claims, together with fail-closed gating when the proven conditions are not met.